

Linear Attention with Global Context: A Multipole Attention Mechanism for Vision and Physics

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Abstract

*Transformers have become the de facto standard for a wide range of tasks, from image classification to physics simulations. Despite their impressive performance, the quadratic complexity of standard Transformers in both memory and time with respect to the input length makes them impractical for processing high-resolution inputs. Therefore, several variants have been proposed, the most successful relying on patchification, downsampling, or coarsening techniques — often at the cost of losing the finest-scale details. In this work, we take a different approach. Inspired by state-of-the-art techniques in n -body numerical simulations, we cast attention as an interaction problem between grid points. We introduce the **Multipole Attention Neural Operator (MANO)** that computes attention in a distance-based multiscale fashion. MANO maintains, in each attention head, a **global receptive field** and has a **linear time and memory complexity** with respect to the number of grid points. Empirical results on image classification and Darcy flows demonstrate that MANO rivals state-of-the-art models, such as ViT and Swin transformer, while reducing runtime and peak memory usage by orders of magnitude. We open-source our code for reproducibility at: <https://github.com/AlexColagrande/MANO>.*

1. Introduction

Convolutional Neural Networks (CNNs) have formed the cornerstone of modern computer vision [22, 29, 31]. Their architectural design leverages the spatial locality and translational invariance properties of images by applying shared convolutional filters over local receptive fields, enabling an efficient parameter usage and a strong inductive bias for grid-structured data.

In recent years, Vision Transformers (ViTs) [16] have emerged as an alternative to CNNs. They are based on the Transformer architecture [54] introduced in the field of Nat-

ural Language Processing (NLP) for sequence-to-sequence learning. This neural architecture is characterized by the use of the self-attention mechanism [2] that allows modeling global contextual information across the *tokens* of a text or the *patches* of an image. Despite lacking the strong locality priors of CNNs, attention-based architectures have demonstrated competitive performance in image classification, particularly when trained on large-scale datasets [44].

Beyond computer vision and NLP, Transformer-based models have found application in scientific machine learning, particularly in the resolution of Partial Differential Equations (PDEs). PDEs constitute the fundamental mathematical framework for modeling a vast array of phenomena across the physical and life sciences - from molecular dynamics to fluid flows and climate evolution. Substantial efforts have been devoted to approximating the solution operators of such equations at scale. Classical numerical solvers — including finite difference [14], finite element [13], and spectral methods [6]— discretize the underlying continuous operators, thereby recasting the problem as a finite-dimensional approximation. More recently, the increasing availability of observational data on structured grids has fostered a paradigm shift towards data-driven approaches such as Physics-Informed Neural Networks (PINNs) [24, 39, 46]. PINNs harness these observations to learn PDE solutions directly, enforcing physical consistency through soft constraints without relying on explicit mesh-based formulations. However, like classical numerical solvers, PINNs are typically designed to approximate the solution of a specific PDE instance — for example, computing the solution corresponding to a fixed coefficient, boundary or initial condition. This means even minor variations in input parameters require re-solving the system or, in the case of neural models, costly re-training. In contrast, operator learning [25] targets a fundamentally more ambitious goal: to approximate a mapping between infinite-dimensional function spaces.

Although considerably more challenging, operator learning offers the advantage to generalize across input condi-

tions without further optimization, offering a scalable and computationally efficient alternative to traditional point-wise solvers. *Note that operator learning is not restricted to PDEs, as images can naturally be viewed as real-valued functions on 2-dimensional domains.*

As in computer vision, recent neural operator models benefit from the development of attention-based architectures [1, 3]. However, attention suffers from quadratic time and memory complexity with respect to the input size, making it impractical for high-resolution data. To tackle this, some variants of the attention mechanism process data in local patches or down-sample the input, drastically cutting computational cost but often sacrificing the fine-grained details crucial for dense-prediction tasks. Other methods replace full attention with low-rank approximations [55], sparsity-inducing schemes [59], or kernel-inspired formulations [12]. Alternatively, Synthesizer proposes to learn attention weights without relying on explicit query–key products [49]. However, these approaches often trade off run-time for expressiveness or ease of implementation.

In this work, we propose an efficient variant of the attention mechanism specifically suited for image classification as well as dense-prediction tasks such as physical simulations. Our method achieves computational gains by relaxing the classical attention formulation while preserving performance by preserving global context.

We propose the **Multipole Attention Neural Operator** (MANO), a novel transformer neural operator in which each head computes attention between a point and a multiscale expansion of the input centered at that point. The attention is performed against a hierarchical decomposition of the input, dynamically downsampled based on the query location. Importantly, we compute the query, key and value matrices Q , K and V at every scale using the same point-wise operator to allow the model to accept inputs at any resolution.

Our contributions are as follows:

- We propose the Multipole Attention Neural Operator (MANO) that formulates attention as an interaction problem and solves it using the Fast Multipole Method.
- By combining MANO with the Swinv2 architecture, we improve transfer learning results on several image classification tasks.
- MANO achieves state-of-the-art results on Darcy flow simulation benchmarks matching, and sometimes surpassing, state of the art baselines.

2. Related work

The Vision Transformer (ViT) [16] was the first to successfully adapt the Transformer architecture to image classification, achieving remarkable performance. It divides the input image into fixed-size patches, flattens them into token embeddings, adds positional encodings, and processes the resulting sequence with a Transformer encoder. When

pretrained on large datasets such as ImageNet-21k [15] or LVD-142M[44], ViTs rival or exceed CNNs on image classification tasks. However, despite their efficiency, they suffer from limited local information interaction, single-feature representation and therefore low-resolution outputs making it sub-optimal for dense prediction tasks.

These limitations have motivated a number of efficient vision transformer variants.

2.1. Efficient Vision Transformer Variants

Swin Transformers The Swin Transformer [35] restricts self-attention to non-overlapping windows that are shifted between layers, yielding hierarchical, multi-scale representations without global attention. Swin Transformer V2 [36] augments this design with learnable-temperature scaled cosine attention, log-spaced relative position bias, and continuous pre-norm, improving high-resolution stability and enabling deeper networks—all while preserving the original’s efficient window-based computation.

Distilled and Compact ViTs : TinyViT [58] uses *pretraining-stage distillation* from a large teacher (e.g., Swin-B/L trained on ImageNet-21k). By caching teacher logits and applying neural architecture search under FLOPs/parameter constraints, TinyViT produces smaller models at only a small performance loss.

Data-Efficient Image Transformers (DeiT) [51] add a learnable *distillation token* that learns from a CNN teacher’s soft logits. Later work [52] adds self-supervised distillation and token pruning for further efficiency.

Collectively, these efforts have greatly extended ViT applicability across resource-constrained tasks. However, the inherent multi-scale structure of images remains only partially integrated into existing alternatives to the attention mechanisms, potentially hindering the overall performance.

2.2. Operator Learning via Multipole Attention

In this work, we illustrate the interest of our proposed multipole attention mechanism for learning solution operators of PDEs directly from input–output pairs, as encountered in tasks like fluid flow estimation and other dense prediction problems [25]. Operator learning was first explored by Lu et al. [37], who established a universal approximation theorem for nonlinear operators using DeepONets, laying theoretical foundations for neural operator approximation. Building on this foundation, the Fourier Neural Operator (FNO) [33] parameterizes an integral kernel in the Fourier domain—using efficient FFT-based convolutions to capture global interactions across the entire domain. These pioneering methods have since inspired a wealth of extensions—but their reliance on global or Fourier-based interactions limits their scalability to very high-resolution grids. These works paved the way for numerous extensions.

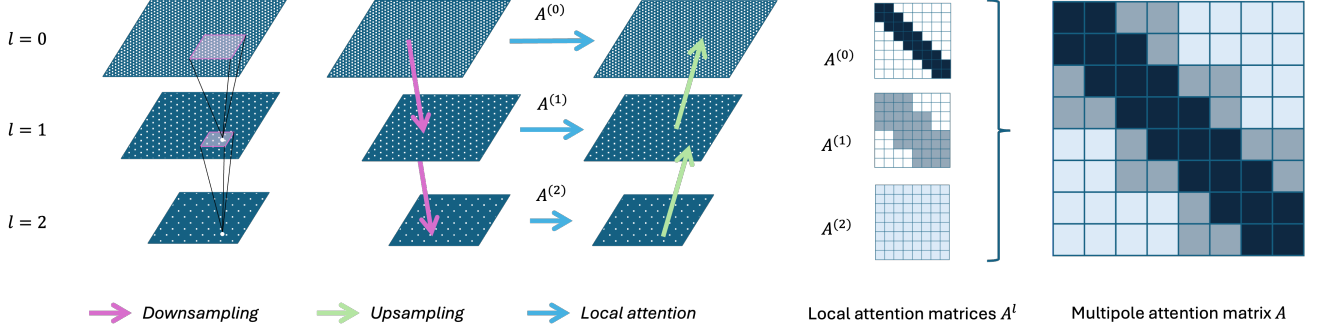


Figure 1. (Left) The multi-scale grid structure. (Center) The V-cycle structure for computing multipole attention with the fast multipole method. (Right) Attention matrices. Illustration with three levels. The attention matrix A is computed in a multiscale manner with respect to each level. The higher the level, the shorter the range of the interaction. At a given layer, down-sampling (resp. up-sampling) is performed using a convolution kernel (resp. deconvolution) shared across all different levels.

Transformer neural operators In [10] the classical transformer was adapted for the first time to operator learning problems related to PDEs. The paper explores two variants, respectively based on the Fourier transform and on the Galerkin method. The latter one uses a simplified attention based operator, without softmax normalization. This solution shares the linear complexity with our work but not the same expressivity. In this line of work, LOCA [34] uses kernel theory to map the input functions to a finite set of features and attends to them by output query location. Recently, [9] proposed to handle attention in a continuous setting and, as well as [56], proposed an operator-learning version of the more classical ViT. Notably, the Universal Physics Transformer (UPT) [1] scales efficiently based on a coarsening of the input mesh.

Multiscale numerical solvers. Our method is inspired by multi-scale numerical solvers [7, 8, 21], in particular the Fast Multipole Method (FMM). A new version of the Fast Multipole Method is introduced by [19] for the evaluation of potential fields in three dimensions and its specialization with the V-cycle algorithm, introduced by [43].

Theoretical studies on transformers. The particle-interaction interpretations of attention was first introduced in Sinkformer [48]. [17] views Transformers as interacting particle systems, they described the geometry of learned representations when the weights are not time dependent, and [18] developed a mathematical framework for analyzing Transformers based on their interpretation as interacting particle systems inspiring us to compute the attention using the most efficient techniques available for solving particle-interaction problems.

Multiscale neural architectures. Several transformer architectures related to the multiscale principle used in our

method were proposed in the one-dimensional setting of Natural Language Processing (NLP) [23, 41, 60, 62], and in graph learning methods [32, 40, 61]

Relation to Fast Multipole Attention Among existing approaches, the closest to ours is Fast Multipole Attention (FMA) [23], which reduces the $O(N^2)$ cost of 1D self-attention via hierarchical grouping: nearby queries attend at full resolution, while distant keys are merged into low-rank summaries, achieving $O(N \log N)$ or $O(N)$.

Our method differs in two key aspects:

- **Input domain.** FMA targets one-dimensional token sequences; we operate on two-dimensional image grids with multiscale spatial windows.
- **Downsampling.** FMA hierarchically downsamples queries, keys, and values. In contrast, we downsample the input feature map *prior* to attention, yielding a self-contained block that integrates seamlessly with standard transformer backbones (e.g., SwinV2) and preserves pre-trained attention weights.

3. Introducing MANO

3.1. Attention as an interaction problem

In this section, we cast the computation of self-attention as a dense n -body interaction problem. An n -body system consists of n entities (often referred to as bodies) whose state is described by a configuration $(x_1, \dots, x_n)$. The evolution of such a system is governed by a set of interaction laws, which in our setting are determined by pairwise interactions specified through a kernel function:

$$\begin{aligned} \kappa : \mathbb{R}^d \times \mathbb{R}^d &\rightarrow \mathbb{R} \\ (x_i, x_j) &\mapsto \kappa(x_i, x_j) \end{aligned}$$

An n -body simulation refers to a numerical method for computing these interactions, typically requiring $O(n^2)$ op-

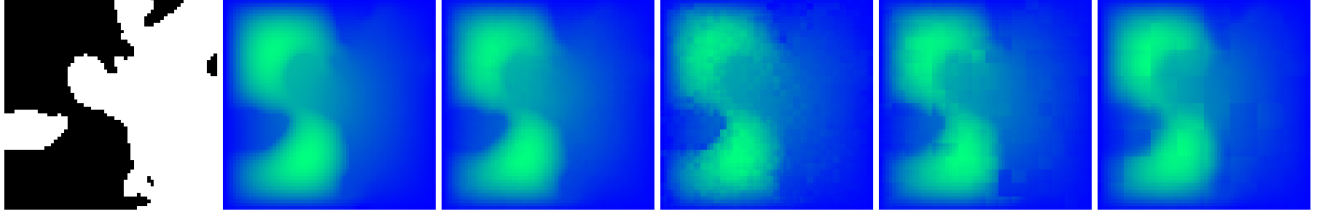


Figure 2. Darcy flow reconstruction: from left to right—input coefficient field, ground truth solution, MANO prediction, and ViT predictions using patch sizes 2, 4, and 8. MANO applies multipole attention using overlapping windows of size 2, and performs downsampling and upsampling across 5 levels using convolutions with kernel size 2×2 , stride 2, and zero padding.

erations due to the dense pairwise structure. This computational cost motivates the development of faster approximations, such as the Fast Multipole Method, which reduces the complexity to $O(n)$.

In the following, we consider a regular grid, corresponding either to the pixels of an image in the classification setting or to the discretized samples of an input function in the operator learning framework. We denote by X a sequence of N observations $(x_1, \dots, x_N)^\top \in \mathbb{R}^{N \times d}$ with elements embedded in dimension d . The self-attention mechanism firstly applies three learnable linear projections to obtain queries, keys and values [54]:

$$Q = XW_q, \quad K = XW_k, \quad V = XW_v \quad (1)$$

with $W_q, W_k, W_v \in \mathbb{R}^{d \times d}$ and $b_q, b_k, b_v \in \mathbb{R}^d$. Next, it computes an $N \times N$ attention matrix A whose i -th row forms a probability distribution over all keys:

$$A_{ij} = \frac{\exp(Q_i^\top K_j / \sqrt{d})}{\sum_{l=1}^N \exp(Q_i^\top K_l / \sqrt{d})}. \quad (2)$$

Finally, each token is updated as a convex combination of the value vectors: $x_i \leftarrow \sum_{j=1}^N A_{ij} V_j$. In this form, one can view the set $\{x_i\}_{i=1}^N$ as a cloud of N particles in $\mathbb{R}^d$ interacting in a pairwise manner via a kernel κ defined as:

$$\kappa(Q_i, K_j) = \exp(Q_i^\top K_j / \sqrt{d}). \quad (3)$$

Therefore, we interpret a self-attention layer as a single time step of a discretized N -body dynamical system. Under this analogy, computing attention is equivalent to predicting the next state of an interacting particle system — and it becomes natural to accelerate this computation using the FMM [19], reducing the usual $O(N^2)$ cost of the pairwise sums to $O(N)$.

3.2. MANO

In this section we detail the Multipole Attention layer as well as the complexity of the model.

Method Overview : Let $X_0 = X \in \mathbb{R}^{H \times W \times d}$ be the original high-resolution image (height H , width W , embedding dimension d). We define L levels of downsampling by a convolutional kernel D with weights shared across levels, producing $X_\ell = D(X_{\ell-1})$ ($\ell = 1, \dots, L$) where $X_\ell \in \mathbb{R}^{H/2^\ell \times W/2^\ell \times d}$. At each level ℓ , we partition the feature map into potentially overlapping sliding windows and, within each window, compute the attention map $A_\ell = \text{Softmax}(Q_\ell K_\ell^\top / \sqrt{d})$ where $Q_\ell, K_\ell, V_\ell \in \mathbb{R}^{(H/2^\ell \times W/2^\ell) \times d}$ are the query, key, and value embeddings extracted from X_ℓ . This restricts self-attention to localized neighborhoods while still enabling cross-window interactions via the sliding overlap and the hierarchical mixing.

We then produce the attended features $\tilde{X}_\ell = A_\ell V_\ell$, and upsample back to the next-finer resolution via a transposed convolution U :

$$\hat{X}_\ell = U(\tilde{X}_\ell) \in \mathbb{R}^{H/2^{\ell-1} \times W/2^{\ell-1} \times d}. \quad (4)$$

Finally, we combine all levels by summation at the original resolution: $X_{\text{out}} = \sum_{\ell=0}^L U^\ell(\text{Attn}(X_\ell))$, where $\text{Attn}(X_\ell)$ denotes $\tilde{X}_\ell$ at level ℓ , and U^0 is the identity.

Sharing the same convolutional kernel for both downsampling and up-sampling—and reusing the same attention weights—keeps the total parameter count constant, regardless of the number of layers L . The convolutions have the role to provide a representation of the input at the next scale, independently of the scale. This ensures that an attention map learned at the finest scale produce effective representations at different scale, even in the case of a pretrained attention from a windowed-attention based model, such as the SwinV2, with finetuning on the convolutional parameters but not on the attention weights. The shared convolutions act as scale-agnostic projectors, producing the next-scale feature representation in the same way at every level. As a result, an attention map learned at the finest resolution remains effective across all scales. In practice, this means one can take a windowed-attention backbone, such as SwinV2, freeze its attention weights, and fine-tune only the convolutional parameters to adapt it to new resolutions without increasing model size.

Computational Complexity. We analyze the cost under non-overlapping windows on a square image of side H (so $N = H^2$ tokens), embedding dimension d , window size w , and down-sampling factor k . The maximum number of levels is $L = \log_k(H)$. We denote by $N_\ell = \frac{N}{k^{2\ell}}$ the number of grid points at level ℓ .

Each windowed self-attention on $M = w^2$ tokens costs $O(M^2d)$ and is applied across $O(N_\ell/M)$ windows, for a total complexity of

$$O(N_\ell M d) \quad (5)$$

Windowed attention computes a standard self-attention for a complexity of $O(M^2d)$ within each of the N/M windows. The total complexity is thus $O(NMd)$. Our Multipole attention iterates the same windowed attention and aggregates the contribution at each level ℓ therefore the total complexity reads

$$\sum_{\ell=0}^L N_\ell M d = \sum_{\ell=0}^L O\left(\frac{N}{k^{2\ell}} M d\right) = O(NMd) \quad (6)$$

So interestingly, even though we apply windowed attention across multiple scales, the total cost remains dominated by the finest-scale pass, with coarser levels adding a negligible additional overhead. As a result, our approach preserves the linear complexity of single-scale windowed attention while delivering significantly greater expressive power.

4. Experimental settings

This section outlines the experimental setup for both image classification and physics simulations.

For image classification, we evaluate on several fine-grained datasets a Swin Transformer V2 modified with our proposed attention mechanism. Models are initialized with weights pretrained on ImageNet-1k. The encoder is frozen (except for the additional convolutions of MANO) and a linear classifier is learnt on the target classification tasks.

For physics simulations, we train all models on instances of the Darcy flow problem, from scratch and across different resolutions.

4.1. Image classification

Datasets. The ImageNet-1k [15] dataset is used to pre-train all models and we perform *linear probing* on several downstream classification benchmarks, namely CIFAR-100[28], Oxford Flowers-102 [42], Stanford Cars [26], Food101 [4], Tiny-ImageNet-202 [30] and Oxford-IIIT Pet Dataset [45].

Architecture. As the backbone for our Multipole Attention model, we adopt the ‘‘Tiny’’ version of the Swin Transformer V2 [36]. Since the attention block is shared across all levels of the multipole hierarchy, MANO can inherit the

pretrained weights from the original Swin Transformer, requiring only a small number of additional trainable parameters: one convolution and one transposed convolution per attention head, along with the classification head. Convolutions have a kernel size of 2 and a stride of 2. This design ensures that our variant introduces a minimal increase in parameter count relative to the base model. Specifically, for the Tiny version, the total number of parameters increases from 27,73M to 28,47M, corresponding to an additional 740,356 parameters, or 2.67% more than the original model.

Training. We freeze the pretrained encoder weights and train a single fully connected layer for 50 epochs on top of the frozen encoder using AdamW as optimizer with a cosine annealing learning rate schedule. Training the models with the original shifted attention of SwinV2 and the models with our proposed multipole attention only differs by a warm-up phase to learn the upsampling and downsampling convolutional filters introduced by MANO. We report the resulting top-1 accuracy of these experiments in Table 1.

4.2. Darcy flow simulation

Task We evaluate our method on the task of steady-state 2D Darcy Flow simulation, a widely used task in the neural operator literature [25]. The problem is based on the following second-order, linear elliptic PDE:

$$-\nabla \cdot (a(x) \nabla u(x)) = f(x), \quad x \in (0, 1)^2, \quad (7)$$

to solve with homogeneous Dirichlet boundary conditions: $u(x) = 0, \quad x \in \partial(0, 1)^2$. In this PDE, the function $a(x)$ represents the spatially varying permeability of a porous medium. The forcing term $f(x)$ is fixed to $f(x) \equiv 1$ across all inputs. The output $u(x)$ is the scalar field representing the pressure within the domain. Although the PDE is linear in u , the map from the input $a(x)$ to the solution $u(x)$ is nonlinear due to the interaction of $a(x)$ with the gradient operator inside the divergence.

The task is to learn this solution operator: given a new input field $a(x)$, the model must predict the corresponding output $u(x)$. In our experiments, $a(x)$ is sampled as a binary field (i.e., values are either 0 or 1), representing a medium composed of two different materials.

Architecture. We use a classical transformer architecture of depth 8 with 4 attention heads per layer. In place of the conventional self-attention, we employ our multipole attention module. Additionally, we apply Layer Normalization at every level to improve training stability and mitigate issues of vanishing or exploding gradients, which can arise due to shared attention across hierarchical levels.

Model	Params.	Complexity	Tiny-IN-202	Cifar-100	Flowers-102	Food-101	StanfordCars-196	OxfordIIITPet
TinyViT [58]	21M	$O(N^2)$	-	75.2%	82.4%	-	<u>61.7%</u>	86.5%
ViT-base [16]	86M	$O(N^2)$	73.07%	<u>80.63%</u>	92.75%	<u>80.31%</u>	41.95%	<u>87.68%</u>
DeiT-small [52]	22M	$O(N^2)$	<u>81.34%</u>	75.15%	66.60%	71.39%	36.38%	<u>87.56%</u>
SwinV2-T [36]	28M	$O(N)$	80.53%	75.47%	56.46%	76.96%	38.36%	87.14%
MANO-tiny	28M	$O(N)$	87.52%	85.08%	<u>89.00%</u>	82.48%	65.68%	88.31%

Table 1. Linear probing accuracies for several image classification datasets. MANO matches or even outperforms the performances of state-of-the-art models. The results for TinyViT are taken from [58], while all other baselines are fine-tuned via linear probing using pretrained backbones¹. For each model, we report the number of parameters and the asymptotic complexity of its attention block with respect to the number of patches N . As a plug-and-play replacement for the attention mechanism, our attention can be applied after patching, similar to Swin, allowing our vision-specific MANO variant to scale linearly with the number of patches.

Training. We train all the considered models for 50 epochs with AdamW optimizer and cosine learning rate scheduler. The initial learning rate is in the order of 10^{-4} . We use the dataset open-sourced in [25], comprised of input-output pairs (a, u) at resolutions $n \times n$ for $n \in \{16, 32, 64\}$. The model is trained to minimize the mean squared error (MSE) on the training set and evaluated on a held-out test set using the relative MSE error, where $\hat{u}$ is the model prediction and u the ground truth solution: $\frac{|\hat{u} - u|_2}{|u|_2}$.

5. Results

5.1. Image Classification Results

Table 1 presents the top-1 accuracy of three models of similar parameter counts (ViT (21M parameters), SwinV2-T (28M), ViT-base (86M), DeiT-small (22M) and MANO (28M)) across six downstream image classification datasets. The reported results for TinyViT are taken from [11, 58] while the results for SwinV2-T are taken from [36].

First, MANO consistently outperforms TinyViT on all benchmarks where results are available. It also surpasses DeiT-small and SwinV2-T when these models are fine-tuned via linear probing using ImageNet-1k pretrained weights, across an expanded set of datasets. Compared with the bigger ViT-base, our model performs better in all the benchmark except than Flowers-102 that is the dataset with less training data among the one considered. the dataset with the smallest training set among those considered. This suggests that in low-data regimes, models with a higher parameter count may have an advantage due to their increased capacity to memorize or adapt to limited supervision.

The improvement ranges from about 1–2 % on easier tasks like Oxford–IIIT Pet to nearly 5/7 points on more challenging datasets such as CIFAR–100 and Tiny-ImageNet. Even compared to ViT-base, which has more than twice the number of parameters, MANO achieves gains of roughly 3–10 points accross all the benchmarks expect for Flowers-102, demonstrating that multiscale hierarchical attention produces significantly more transferable features without increasing too much model size.

Second, the advantage of MANO becomes especially pronounced on fine-grained classification tasks. On Flowers–102 and Stanford Cars, SwinV2-T achieves only 56.5% and 38.4% accuracy, respectively, while TinyViT recovers to 82.4% and 61.7%. In both cases, MANO further improves performance to 89.0% on Flowers–102 and 65.7% on Cars, indicating that combining local details (e.g., petal shapes or headlight contours) with global context (overall flower appearance or car silhouette) is critical for distinguishing highly similar classes.

Third, on medium-difficulty datasets such as Tiny-ImageNet–202 and CIFAR–100, MANO again holds a clear lead. It outperforms SwinV2-T by approximately 7 points on Tiny-ImageNet–202 and by around 10 points on CIFAR–100. These results suggest that attending to multiple resolutions—capturing both fine textures and broader scene structures—yields better representations than the single-scale windowed attention used in SwinV2-T.

Finally, although MANO and SwinV2-T share the same parameter count (28M), MANO delivers a consistent 5–10 point advantage on mid-level benchmarks and maintains a smaller lead on easier tasks like Oxford–IIIT Pet. TinyViT’s 21M parameters are insufficient to match either 28M model, underscoring that hierarchical multiscale attention makes more effective use of model capacity than either pure global self-attention (ViT) or fixed-window local attention (SwinV2-T).

5.2. Darcy Flow Simulation Results

Model	16×16	32×32	64×64
FNO	0.0195	0.0050	0.0035
ViT patch_size=8	0.0160	<u>0.0038</u>	0.0021
ViT patch_size=4	0.0179	0.0039	<u>0.0019</u>
ViT patch_size=2	0.0169	0.0049	0.0026
Local Attention	<u>0.0133</u>	0.0188	0.0431
MANO	0.0080	0.0020	0.0013

Table 2. Benchmark on Darcy flow simulations. Relative MSE. A given model is evaluated and tested on the same resolution, either 16×16 , 32×32 or 64×64 .

Table 2 presents the relative MSE of various models trained from scratch on Darcy flow at different resolutions. The Fourier Neural Operator (FNO) [33] is a neural operator designed to learn mappings between function spaces, such as the coefficient-to-solution map for PDEs. By operating in the Fourier domain, the FNO captures long-range dependencies across the entire domain with near-linear complexity $O(N^2 \log N)$ (for an $N \times N$ grid), making it effective for a wide range of PDE-based tasks, leading to state-of-the-art results on tasks such as Darcy Flow Simulation. It achieves MSEs of 0.0195, 0.0050, and 0.0035 as the grid is refined, showcasing its strength in capturing global spectral components but its limited ability to resolve fine-scale details at coarser resolutions.

For ViT, we evaluate patch sizes of 8, 4, and 2: the patch-4 variant attains the best errors (0.0179, 0.0039, 0.0019), whereas smaller patches (size 2) slightly worsen performance at low resolution and fail to match patch-4 at 64^2 . This sensitivity indicates that vanilla ViT’s pure global attention is capable of approximating the solution operator but depends heavily on patch granularity. In contrast, a pure local-attention model (fixed window) degrades sharply at 32^2 and 64^2 , since local windows cannot propagate long-range dependencies across the domain.

By combining fine-grid attention (to capture local conductivity channels) with progressively coarser resolutions (to model global pressure fields), MANO consistently achieves the lowest errors, roughly halving the MSE of both FNO and standard ViT at every scale and overcoming the locality limitations inherent in fixed-window attention.

Lastly, we examine in Figure 2 the quality of the reconstructed images, demonstrating that MANO’s multi-scale modeling recovers both sharp transitions and smooth boundaries with high fidelity, even on a coarse grid. By contrast, images reconstructed by ViT exhibit noticeable patching artifacts.

In summary, MANO’s multiscale hierarchical attention achieves state-of-the-art performance on both Darcy flow simulations and image classification tasks. Its design makes it well suited to the corresponding data, as it captures fine-scale detail and broad-scale context simultaneously.

Hyperparameters. A detailed table of hyperparameters is provided in the Appendix; below, we outline our main design choices.

The “number of levels” specifies how many hierarchical scales are included in the multipole attention; with a window size of 8 for the windowed attention, we achieve

¹Checkpoints are available at <https://huggingface.co/google/vit-base-patch16-224> (ViT-base), <https://huggingface.co/facebook/deit-small-patch16-224> (DeiT-Small), and https://huggingface.co/timm/swinv2_tiny_window8_256.ms_in1k (SwinV2-T).

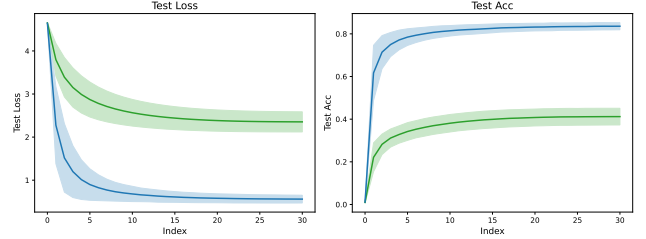


Figure 3. Ablation study comparing average pooling (green) and learnable convolutions (blue) for the sampling step in MANO. We report the Cross-Entropy validation loss (left) and accuracy (right) on CIFAR-100. Mean and standard deviation over fifteen runs are reported, varying the learning rate between 10^{-3} and 10^{-4} .

the best performance using the maximum of 3 levels. Due to Swin’s built-in downsampling between stages, this corresponds to 3 levels across the first two layers, 2 levels for the next two layers, and a single level for the remaining eight layers. To coarsen the input grid, we compared average pooling versus learned convolutions—convolutions consistently outperformed pooling. For upsampling, transpose convolutions outperformed nearest-neighbor interpolation. When used, `kernel_size` and `stride` refer to these convolutional operations.

As shown in Figure 3, using learned convolutions for both down- and up-sampling significantly improves expressivity: even with pretrained attention weights, convolution-based sampling enables a windowed attention trained at one resolution to transfer effectively to another. Note that a single convolutional kernel is reused for all downsampling operations, and a separate kernel is reused for all upsampling operations.

6. Discussion

Hierarchy depth in vision vs. Physics. In our image classification experiments, we follow SwinV2-T’s architecture and compute attention at three hierarchical levels in early stages, two in the middle, one at the end of the encoder). For Darcy flow grids, we set downsampling levels so that the coarsest scale is 2×2 , the smallest possible when using a 2×2 window for the attention. We found that increasing the number of levels consistently improves performance, however we note that in physics simulation, the number of levels can be treated as a hyperparameter and tuned based on input resolution and the desired balance between local and global interactions.

Limitations. MANO’s current design uses a fixed, static hierarchy and attention parametrization. While effective, it could benefit from learnable scale selection and explicit cross-level interactions to better capture multi-scale couplings. Additionally, we assume a uniform grid to discretize

the input. This simplifies implementation but fails to capture regions with steep gradients or intricate boundaries in physics simulations. Introducing adaptive meshing would thus improve accuracy and efficiency for localized phenomena that otherwise demand a higher resolution simulation, but would require redefining attention over nonuniform spatial supports. Finally, our current implementation could be accelerated by a hardware—optimized GPU kernel to further reduce runtime.

Future Directions. Beyond steady-state Darcy flow, MANO can naturally extend to time-dependent PDEs by integrating with recurrent timestepping or operator-splitting schemes, preserving its ability to capture both spatial multiscale structure and temporal evolution. We also plan to extend our method to unstructured meshes and irregular domains common in real-world physics simulations. On the computer-vision side, applying MANO to dense prediction tasks—such as semantic segmentation or image inpainting—is promising, since its multiscale attention could dynamically balance local details and global context more effectively than standard U-Net architectures.

7. Conclusion

We propose **MANO**, an efficient attention-based architecture inspired by n -body methods, which interprets attention as interactions among mesh points. By introducing a distance-based, multiscale attention mechanism, MANO achieves linear time and memory complexity per head while preserving a global receptive field. Across several image classification benchmarks and Darcy flow simulations, MANO matches the accuracy of full-attention models yet substantially reduces runtime and peak memory. Unlike patch-based approximations, it avoids discontinuities and retains long-range dependencies inherent to physical systems. Our results demonstrate that MANO is a scalable alternative for both vision tasks and mesh-based simulations. Future work includes applying MANO to dense vision tasks such as semantic segmentation, and to extend it to irregular meshes and a broader class of physical simulations.

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